

Partial Derivatives as Engineering Sensitivities

A visual supplement to AS1109 Lessons 4–5

Section H – Calculus and Ordinary Differential Equations

Central idea. If an engineering quantity $z = f(x, y)$ depends on two independent inputs, then

$f_x(a, b)$ measures the local rate of change of z when x changes and y is held fixed,

while $f_y(a, b)$ does the same for changes in y with x held fixed. A partial derivative is therefore a **local sensitivity coefficient**. Its sign tells the direction of change; its magnitude tells how strongly the output responds per unit change in that input.

1. Revisiting the temperature example carefully

The class example was

$$T(x, y) = 100 - x^2 - 2y^2.$$

Think of T as the temperature of a thin plate, with x and y giving position on the plate. If x, y are measured in metres and T in $^{\circ}\text{C}$, then the partial derivatives have units $^{\circ}\text{C}/\text{m}$.

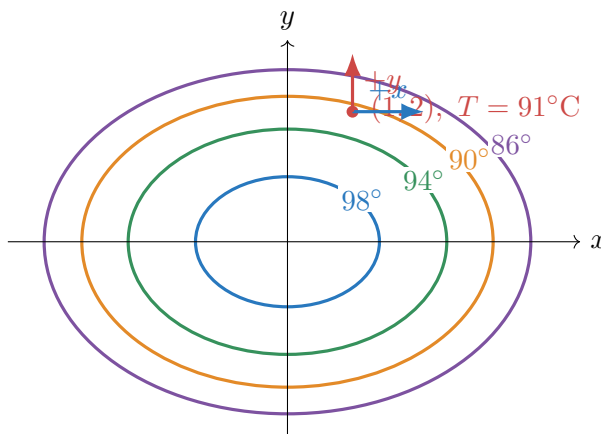
$$T_x = -2x, \quad T_y = -4y.$$

At the point $(1, 2)$,

$$T(1, 2) = 100 - 1 - 8 = 91^{\circ}\text{C},$$

and

$$T_x(1, 2) = -2, \quad T_y(1, 2) = -8.$$



Temperature contours: moving outward crosses lower-temperature contours.

What does $T_x(1, 2) = -2$ mean? Freeze $y = 2$. Then

$$T(x, 2) = 92 - x^2.$$

Near $x = 1$, increasing x by one metre would initially change temperature at about -2°C per metre. For a much smaller movement, say $\Delta x = 0.10$ m,

$$\Delta T \approx T_x(1, 2)\Delta x = (-2)(0.10) = -0.20^{\circ}\text{C}.$$

The exact change is

$$T(1.1, 2) - T(1, 2) = 90.79 - 91 = -0.21^{\circ}\text{C},$$

which is close because the step is small.

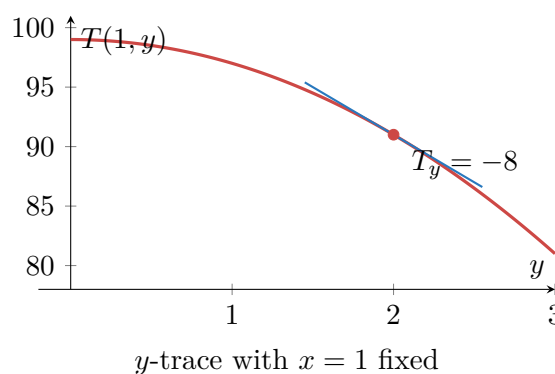
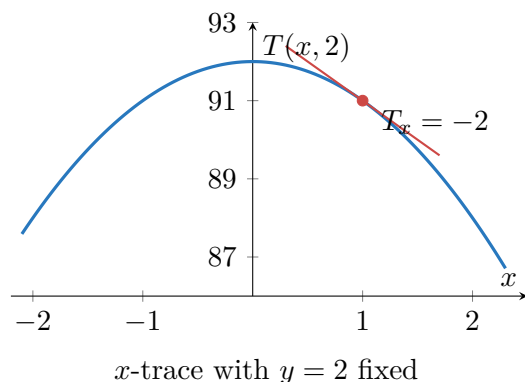
What does $T_y(1, 2) = -8$ mean? Freeze $x = 1$. Then

$$T(1, y) = 99 - 2y^2.$$

For a small movement $\Delta y = 0.10$ m,

$$\Delta T \approx T_y(1, 2)\Delta y = (-8)(0.10) = -0.80^\circ\text{C}.$$

The exact change is $T(1, 2.1) - T(1, 2) = -0.82^\circ\text{C}$.



Engineering interpretation. At $(1, 2)$ the temperature decreases in both positive coordinate directions, but it is locally four times as sensitive to a metre of y -motion as to a metre of x -motion. In thermal design, such sensitivities help identify where temperature changes most sharply and where sensors, insulation, or cooling may need closer attention.

Three cautions. (1) -2 and -8 are *local* rates, not constant rates everywhere. (2) They describe only the coordinate directions. Rates in arbitrary directions are handled later using directional derivatives/gradients. (3) Magnitudes can be compared directly only when the input coordinates use compatible units/scales.

2. Electrical engineering: power sensitivity in a resistor

For a resistive load,

$$P(V, R) = \frac{V^2}{R},$$

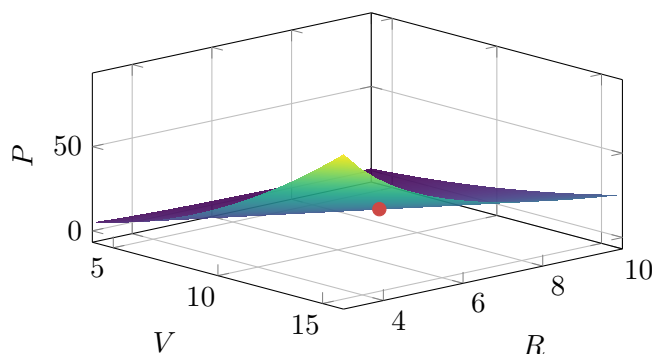
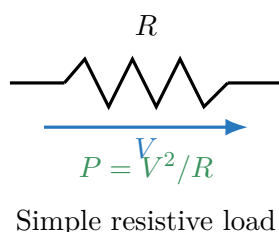
where P is power, V voltage and R resistance. Holding one input fixed at a time,

$$P_V = \frac{2V}{R}, \quad P_R = -\frac{V^2}{R^2}.$$

At $V = 12$ V and $R = 6\ \Omega$,

$$P = 24\ \text{W}, \quad P_V = 4\ \text{W/V}, \quad P_R = -4\ \text{W}/\Omega.$$

Thus, near this operating point, an extra 0.1 V raises power by about 0.4 W if R stays fixed, whereas an extra $0.1\ \Omega$ lowers power by about 0.4 W if V stays fixed.



Where this matters. Partial derivatives quantify operating-point sensitivity in circuit design. They tell an engineer how much power changes because of voltage variation versus component-tolerance variation, while deliberately freezing the other input.

3. Mechanical engineering: cantilever-beam deflection

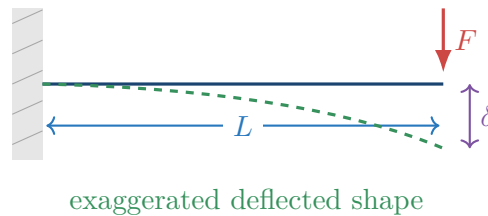
For an ideal cantilever beam carrying an end load F , a standard small-deflection model for the tip displacement is

$$\delta(F, L) = \frac{FL^3}{3EI},$$

where L is beam length, E Young's modulus and I the second moment of area. Treat E and I as fixed material/geometry parameters. Then

$$\delta_F = \frac{L^3}{3EI}, \quad \delta_L = \frac{FL^2}{EI}.$$

The first derivative says deflection responds linearly to load. The second shows a much stronger dependence on length because the model contains L^3 .



For example, with $F = 1000$ N, $L = 2$ m, $E = 200$ GPa and $I = 8 \times 10^{-6}$ m⁴,

$$\delta \approx 1.67 \text{ mm}.$$

At this operating point,

$$\delta_F \approx 0.00167 \text{ mm/N},$$

so a 10 N increase in load changes the predicted tip deflection by about 0.0167 mm. Also,

$$\delta_L = 0.0025 \frac{\text{m}}{\text{m}},$$

so increasing the length by 1 cm predicts an additional deflection of about 0.025 mm to first order.

Where this matters. Design sensitivities tell structural/mechanical engineers which parameters strongly influence deflection. They are useful in tolerance studies, lightweight design and deciding whether changing geometry or reducing load will have the larger local effect.

4. Instrumentation: cross-sensitivity of a sensor

Real sensors often respond to more than the quantity they are intended to measure. Suppose a calibrated output (in mV) depends on temperature T (in °C) and pressure p (in kPa):

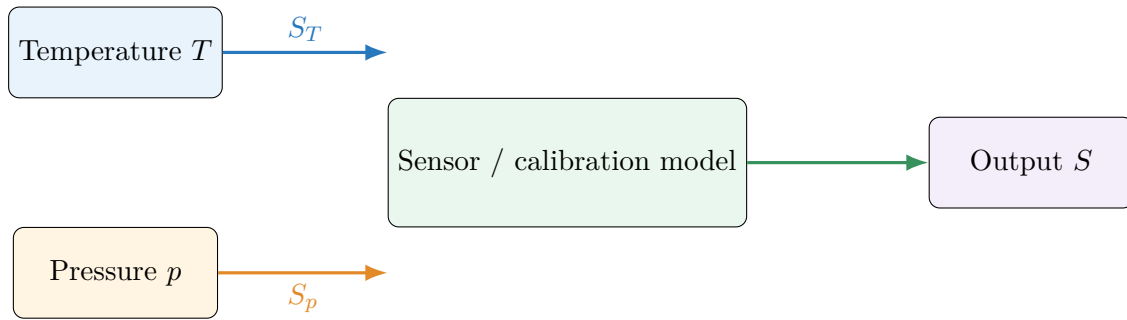
$$S(T, p) = 0.020T + 0.001p + 10^{-5}Tp.$$

Then

$$S_T = 0.020 + 10^{-5}p, \quad S_p = 0.001 + 10^{-5}T.$$

At $T = 50^\circ\text{C}$ and $p = 100$ kPa,

$$S_T = 0.021 \text{ mV}/^\circ\text{C}, \quad S_p = 0.0015 \text{ mV/kPa}.$$



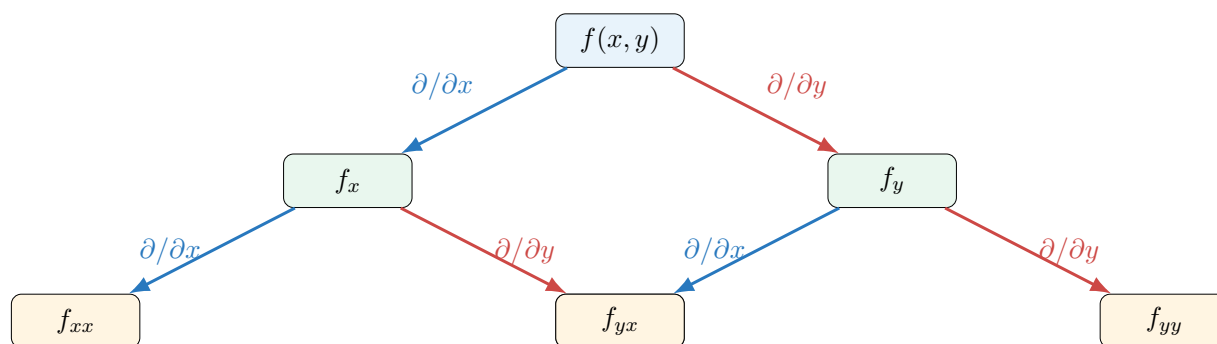
Where this matters. S_T is the local temperature sensitivity when pressure is fixed; S_p is the pressure sensitivity when temperature is fixed. Instrumentation engineers use such derivatives to quantify cross-sensitivity, calibrate sensors, and decide which environmental variable requires compensation.

5. Higher-order partial derivatives: the standard examination pattern

The prescribed next step is to differentiate the first partial derivatives again. For a function $f = f(x, y)$, the four second partial derivatives are

$$\begin{aligned} f_{xx} &= \frac{\partial}{\partial x}(f_x), & f_{xy} &= \frac{\partial}{\partial y}(f_x), \\ f_{yx} &= \frac{\partial}{\partial x}(f_y), & f_{yy} &= \frac{\partial}{\partial y}(f_y). \end{aligned}$$

Thus f_{xx} and f_{yy} are *pure* second partials, while f_{xy} and f_{yx} are *mixed* second partials.



A reliable calculation route: first find f_x, f_y , then differentiate each once more.

Solved textbook-style example

Consider the standard form used in the Lesson 5 notes:

$$f(x, y) = x^3y^2 + e^{xy}.$$

First derivatives:

$$f_x = 3x^2y^2 + ye^{xy}, \quad f_y = 2x^3y + xe^{xy}.$$

Differentiate again:

$$\begin{aligned} f_{xx} &= 6xy^2 + y^2e^{xy}, \\ f_{xy} &= 6x^2y + e^{xy} + xye^{xy}, \\ f_{yx} &= 6x^2y + e^{xy} + xye^{xy}, \\ f_{yy} &= 2x^3 + x^2e^{xy}. \end{aligned}$$

Hence, in this smooth example,

$$f_{xy} = f_{yx}.$$

Exam habit. Show f_x and f_y before the second partial derivatives. This makes the working easy to follow and reduces sign and product-rule errors. For terms such as e^{xy} , explicitly apply the chain rule at each differentiation.

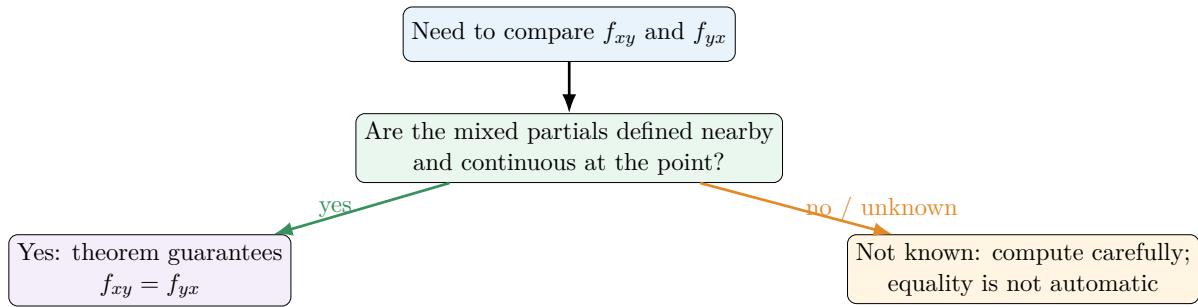
6. Mixed Derivative Theorem

The prescribed theorem explains when the two mixed partials are guaranteed to agree.

Mixed Derivative Theorem. If f_x, f_y, f_{xy}, f_{yx} are defined throughout an open region containing (a, b) and the mixed partials are continuous at (a, b) , then

$$f_{xy}(a, b) = f_{yx}(a, b).$$

Do not replace this by the unconditional statement “mixed partials are always equal.”



7. Engineering application of mixed partials: sensor cross-sensitivity

Return to the illustrative sensor model from Section 4:

$$S(T, p) = 0.020T + 0.001p + 10^{-5}Tp.$$

The first partials were

$$S_T = 0.020 + 10^{-5}p, \quad S_p = 0.001 + 10^{-5}T.$$

Now differentiate those sensitivities:

$$S_{TT} = 0, \quad S_{Tp} = 10^{-5}, \quad S_{pT} = 10^{-5}, \quad S_{pp} = 0.$$

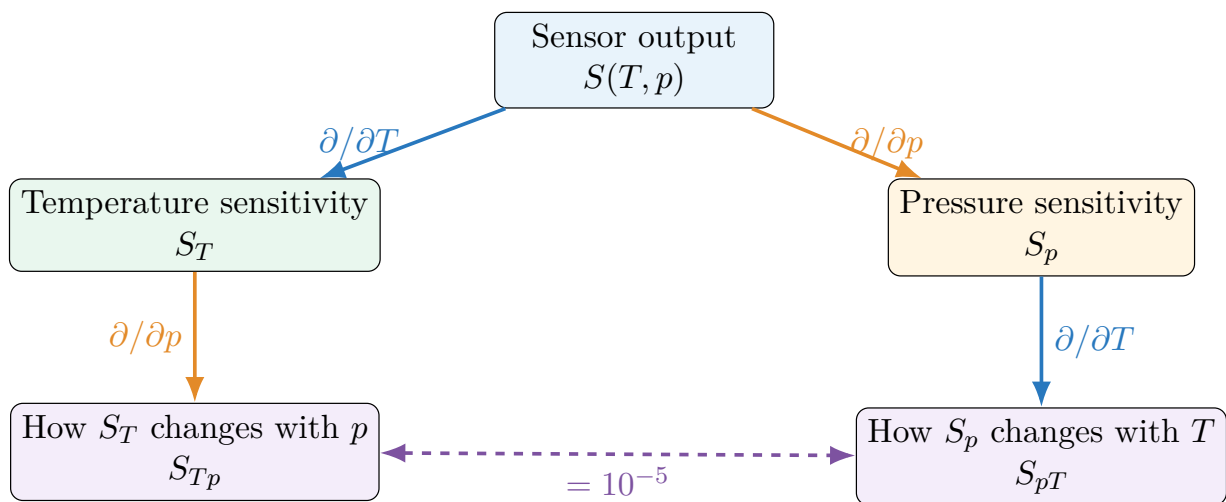
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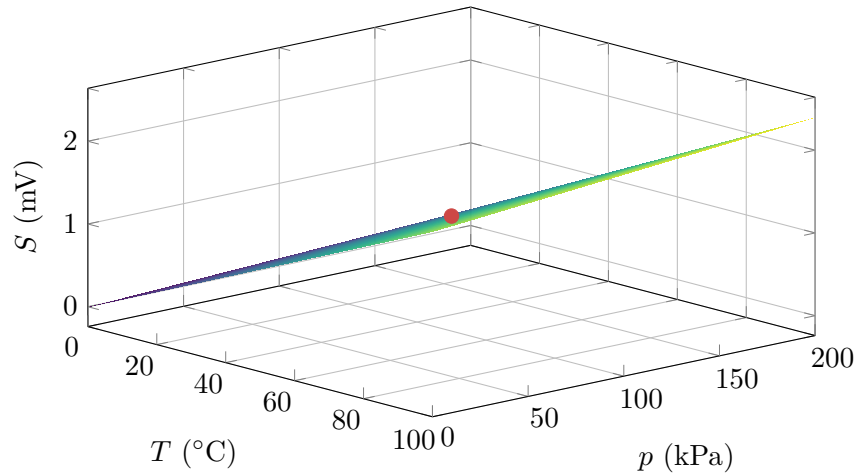
$$S_T = 0.020 + 10^{-5}p,$$

we have

$$S_{Tp} = \frac{\partial}{\partial p}(S_T) = 10^{-5}.$$

Thus the *temperature sensitivity itself* increases by 10^{-5} mV/°C for each additional kPa of pressure. Equivalently, because S_{pT} has the same value, the pressure sensitivity changes at the corresponding rate as temperature changes.





The bilinear Tp term makes the local slope in one input depend on the other input.

Engineering meaning. Mixed partial derivatives are natural measures of *cross-sensitivity*: they quantify how the sensitivity to one input changes when a second input changes. In instrumentation this matters when an intended measurement channel is affected by environmental variables such as temperature, pressure or humidity.

A second, very simple thermal example

For the class temperature model

$$T(x, y) = 100 - x^2 - 2y^2,$$

$$T_{xx} = -2, \quad T_{xy} = 0, \quad T_{yx} = 0, \quad T_{yy} = -4.$$

Here the zero mixed partials say that the x -direction sensitivity $T_x = -2x$ does not depend on y , and the y -direction sensitivity $T_y = -4y$ does not depend on x . The unequal pure second partials, -2 and -4 , show that the two coordinate traces have different curvature.

8. A compact interpretation checklist

Whenever you compute a partial derivative in an engineering model, ask:

1. **What is being changed?** That variable is the differentiation variable.
2. **What is being held fixed?** Every other independent input.
3. **What are the units?** Output units divided by input units.
4. **What does the sign mean?** Positive means local increase; negative means local decrease in the positive coordinate/input direction.
5. **What does the magnitude mean?** Local sensitivity per unit change at that operating point.
6. **Is the statement local?** Yes. The derivative generally changes from point to point.

Bridge to the next lesson. First partial derivatives tell us how the output changes. In the next step we differentiate those sensitivities again. The quantities $f_{xx}, f_{xy}, f_{yx}, f_{yy}$ describe how first-order sensitivities themselves change. These second partial derivatives lead naturally to curvature, mixed effects and important engineering PDEs.

9. Quick check

For the temperature field $T = 100 - x^2 - 2y^2$:

$$T_{xx} = -2, \quad T_{xy} = 0, \quad T_{yx} = 0, \quad T_{yy} = -4.$$

Interpretation: the x -direction temperature slope becomes $2^\circ\text{C}/\text{m}$ steeper in the negative direction for each extra metre of x ; the corresponding y -rate changes twice as fast. The zero mixed partials mean that, in this deliberately simple model, the x -sensitivity does not depend on y and the y -sensitivity does not depend on x .

Standard exam-style practice

1. For $f(x, y) = x^2y + xy^2$, find all four second partial derivatives.

$$f_x = 2xy + y^2, \quad f_y = x^2 + 2xy,$$

so

$$\boxed{f_{xx} = 2y, \quad f_{xy} = 2x + 2y, \quad f_{yx} = 2x + 2y, \quad f_{yy} = 2x.}$$

2. For $f(x, y) = e^{xy}$, compute the mixed partial derivatives.

$$f_x = ye^{xy}, \quad f_y = xe^{xy},$$

therefore

$$\boxed{f_{xy} = e^{xy} + xye^{xy} = f_{yx}.$$

For examination preparation. Prioritise the standard symbolic tasks: compute first and second partial derivatives correctly, distinguish pure and mixed partials, and state the Mixed Derivative Theorem with its hypotheses. The engineering examples in this supplement are intended to make the same standard mathematics easier to understand and remember.